Supporting Information (SI)

Three Centuries of the Laws of Cricket Reveal Core Principles of the Evolution of Regulatory Mechanisms

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Extended Materials and Methods

The Laws of Cricket Data Set

We collected all forty-four editions of the Laws of Cricket, spanning 1752-2019 (268 years), from the library at the Marylebone Cricket Club at Lord's Cricket Ground in London. All rule books that were available from the archives were scanned with a CZUR ET18 Pro Professional Document Scanner, with built-in auto-flattening software. These editions were scanned at the premises (over a three day period in September 2024), where optical-character recognition (OCR) software was then used to convert the scanned documents into machine-readable text. We cleaned this text with Google's Gemini AI. From the AI-cleaned texts, we then extracted the word statistics, hierarchical relationships between the rules (tree structures), and interdependencies between the rules as measured by the interdependency networks between Laws in the rule books. A summary of this process from data collection to construction of the processed data is given in Fig. S1. For more details, please refer to the Materials and Methods. For access to the data in raw, text-corrected, and processed forms, please see the *Data Availability Statement* below at the end of the text.

Processing of the Raw Cricket Rule Books

Having all the scanned PDF files of the raw cricket rule books, the first step was to decide which version of the yearbook to keep when multiple versions existed for the same year. There were different publications that we found—some containing rules for a single year, some containing rules for multiple years—and we had to determine which version to use as the data point for that year in our analysis. We thus developed heuristics to make this determination, such as whether the rule set was a primary source (as opposed to part of a compilation), as well as whether its contents and structure made sense chronologically given the rule sets that came before and after it.

The next step was to decide which sections of the rule set to keep for our analysis. Over these centuries, these rule sets often included some additional segments: ranging from segments on the dimensions of the balls or wickets, to segments on Laws for specific countries, to segments on rules of county cricket that centered more on how players can qualify for certain competitions. We developed heuristics to determine which segments to keep, such as prioritizing rules related to gameplay, as well as factoring in whether the rules were in the main enumerated rule set or in additional segments at the end. See the Data Appendix (data_appendix.xlsx) in the GitHub repository for a detailed description of the aforementioned heuristics.

After finalizing the rule set version for each year and the contents to keep for each of them, we did a manual check of the PDF files to check for strange anomalies (e.g., missing pages, pages wrongly ordered, errors in the Optical Character Recognition output). In most cases, this just involved deleting unnecessary segments; in rare instances where there was missing content, we had to look at the trends of the corresponding segments in the preceding and following rule sets to interpolate the missing content contained, and imputed it into the text file.

The final step was to use the Gemini API to do the remaining cleaning, with the following prompt: "Please correct the following OCR-extracted text by fixing typos, spacing, and punctuation. Ensure that the cleaned text remains true to the original meaning without adding or removing content. Return only the cleaned text without additional comments. Original text: text".
This gave us machine-readable `.txt` files with the text for each of the years with cricket rule sets, that we would carry on to use for constructing the rule book trees and interdependency networks. The Gemini API

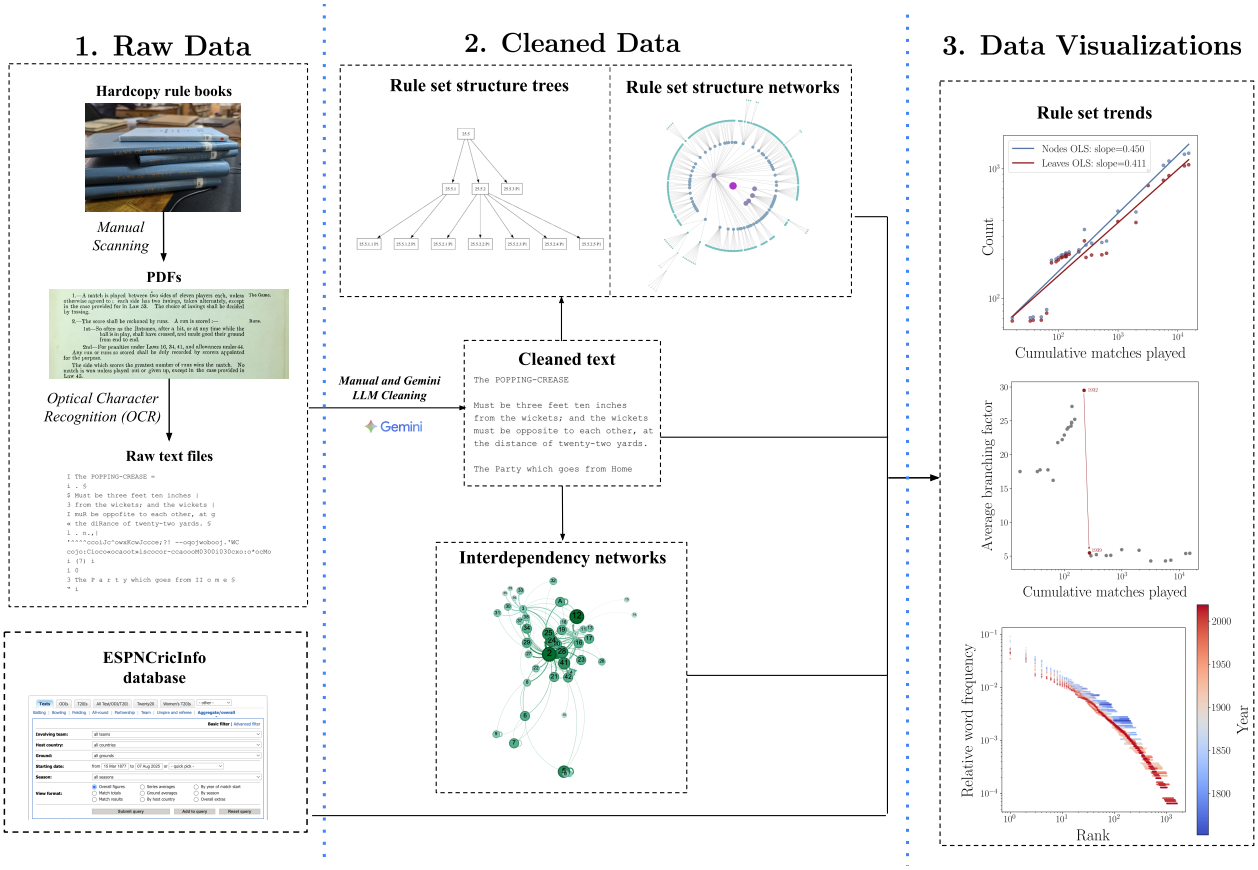

Fig. S1. Flow chart depicting the steps in the data pipeline. We cleaned the raw text data and derived two alternate representations of the cleaned text data: (i) the rule set structure trees/networks, and (ii) the interdependency networks. Finally, we visualized the trends for various metrics of these different representations of the cricket rule set.

corrected two main types of errors: typos due to specific letters in the scanned text being distorted when read in by Optical Character Recognition, and typos due to Optical Character Recognition reading in ornamental elements of the page (page decorations, borders around the text) as actual symbols and letters. A summary of this processing procedure is provided in Fig. S1. The correction of both the aforementioned types of errors can be seen in the difference between the screenshots titled "Raw text files" and "Cleaned text" in the figure.

Fitting Zipfian Distributions to the Rule Set Data

We first used the `nltk` Python package to clean and tokenize the text of each rule set. Next, we obtained the normalized rank-frequency distribution of the words in each rule set, using the relative frequencies of words rather than raw counts. We then fit a Zipf-Mandelbrot curve $f(r) \propto (r+\beta)^{-\alpha}$ to this rank-frequency distribution Piantadosi [2014], and obtained the parameters α and β along with R^2 goodness-of-fit. We plot the exponent α over time in Fig. 2(d), and we show all Zipf-Mandelbrot fits in Figs. S4 and S5.

Making the Rule Book Trees from Document Structure

Different years of the rule books had different numbering schemes to codify their document structure. For example, the 1980 rule set was enumerated via a combination of numbers, letters and roman numerals to indicate different levels of depth in the rule set (e.g. "8(a)(i)"), while the 2019 rule set was enumerated via a decimal system (e.g. "19.1.1.7"). In order to extract their rule set structure trees, we needed algorithms that used regular expression matching to identify the aforementioned expressions signifying enumeration. However,

these differences in numbering schemes meant that different algorithms were required for each.

Fortunately, rule books from the same time period often shared a numbering scheme. We wrote different algorithms for different time periods of rule books grouped by the numbering schemes they used, and automatedly extracted the rule set structure for the whole dataset. This structure was saved in the `.yaml` file format, a data format that allows for easy storage of hierarchical data.

After that, we used the `anytree` Python package to convert the `.yaml` file data into a `.html` visualization of the tree structure. As a correctness check, for each of the `.html` visualizations, we did a comparison with the original PDF of the rule set to check if there were discrepancies. For examples, please see Fig. 4. This was first used to identify and rectify issues with the algorithms used to extract the rule set structures from the rule set text files. Often, however, there were minor errors in the cleaned text files themselves (generated by Optical Character Recognition and processed through Gemini) that distorted the expressions used to identify enumeration and made the algorithm unable to detect them. In these instances, we rectified the issues in the text files themselves before re-generating the tree.

Although visualizing the rule set structures as trees was intuitive, one limitation was that the trees were extremely wide horizontally, therefore tree visualization was only helpful when zooming in on a particular segment of the tree. In other words, it was difficult to gain any insight from looking at the tree as a whole. To address this, we generated a network visualization of each tree in a radial structure: generating visualizations separately with the `kamada-kawai` layout and the `twopi` layout from `NetworkX`.

Lastly, one degree of freedom we introduced was a distinction between "original" and "flattened" rule set structure trees. We were not interested in merely how deep the tree is purely in terms of syntax: instead, we wanted to use the syntactic depth of the tree as a proxy for the degree of specificity of rules. In our original tree configuration, enumerating the paragraphs with numbers added an extra layer of depth to the tree, causing a jump in average depth at around 1828. However, merely tacking a number onto each paragraph was extremely unlikely to correspond to an increase in specificity of the rules. Hence, we decided to create a "flattened" version of the rule set, that places rule paragraphs on the same level of depth of the tree as the enumerations they correspond to, rather than having an extra level of depth for the paragraphs. This would ensure that our measure of the tree depth syntax-wise would actually correspond to the quantity we were interested in measuring, the degree of specificity of the rules.

Making the Interdependency Networks

Different years of the Laws of Cricket recorded citations in different ways. Initial attempts to use Generative AI to detect citations were unsuccessful, hence we resorted to using regular expression matching. Based on spot checks on the generated citation lists across multiple years, comparing them against the actual ruleset PDFs, we estimate an accuracy rate of around 90-95% for our approach.

First, we adapted the code from parsing the document structures, and used it to match every rule in the rule set with the rule text that it contained. Next, we used regular expression matching to identify citations in each block of rule text, recording both the citing rule and the rule being cited. We then aggregated these citations at the Law level, recording the total citations from rules in one law to rules in another law. After that, we constructed a directed, weighted graph in `NetworkX` where the nodes represented the Laws, and the weights of the directed edges were the total citations from the Law represented by the source node to the Law represented by the target node. Lastly, we used network visualization software Gephi to visualize the network, determining the size and color of the nodes based on their degree, for examples please see Fig. 6(top row).

Model-Fitting to In- and Out-Weight Distributions

After generating and visualizing the interdependency networks, we moved on to our analysis of the networks. We obtained the in-degree, out-degree, in-weight and out-weight distributions for each year's interdependency network. Then, we fit Poisson, power-law, and power-law with exponential cutoff distributions to the observed

data, which have the respective functional forms:

$$P_1(x) = \lambda^x e^{-\lambda}/x!, \tag{1}$$

$$P_2(x) \sim x^{-\alpha}, \tag{2}$$

$$P_3(x) \sim e^{-\lambda x} x^{-\alpha}, \tag{3}$$

where the latter two distributions are specified up to their normalization constants. We used the Akaike-information criterion (AIC) to choose the best fit between the proposed functional form and the data. This ensures that the two-parameter power-law distribution with exponential cutoff does not get an unfair advantage over the single-parameter power-law and Poisson distributions.

It is worth noting that we fit all our models on strictly positive support values—nodes with degree/weight equal to 0 were excluded from consideration. The reason for this was twofold: (1) power-law distributions are undefined at $x = 0$, which makes it impossible to include nodes with degree/weight equal to 0 in the fitting process, and (2) we can reasonably consider nodes with 0 degree or weight to not be a part of the interdependency network, since they are neither cited nor contain citations. For consistency, we wanted to make sure all distributions were fit on the same support, hence these nodes were removed for all three distributional fits. Additionally, we normalized distributional fits over their observed support, rather than their full theoretical domain (i.e., the positive integers). This ensures that all three distributions were evaluated as conditional distributions over the same support, making their AIC scores directly comparable.

Eigenvector Centrality

For the eigenvector centrality, we computed eigenvector centrality both with and without edge weights factored in. The rankings of the Laws (top-five and individual case studies) were constructed based on eigenvector centrality with edge weight factored in. There were a couple of complications for the labeling of Laws with their eigenvector centrality. The first was that the Law number of a Law changes over time. For example, the Law "Caught" is Law 22 in 1896, Law 35 in 1947, Law 32 in 1980 and Law 33 in 2019. Additionally, not all Laws were stable over time. For example, over the lifespan of the Laws of Cricket, "The Fieldsman" was renamed to "The Fielder", "Handled the ball" was subsumed into "Obstructing the field", "Appointment of Umpires" was subsumed into "The Umpires". These factors combined meant that there was no clean automated way to track where a single Law was in the rule sets over time. Hence, all labeling had to be done manually, by looking through the rule set PDFs and identifying the rule of interest.

We constructed the table of the top five ranked Laws over time. For each year, we took the eigenvector centrality of the nodes in the interdependency network and ranked them in decreasing order. We then took the Law numbers of the top five Laws and searched them up in the rule set PDFs to identify the name of each Law. We did this for all years of the rule sets that we had citation data for. While each of the Laws in later years had a clear title, there were instances in earlier rule sets where either (1) the title would contain multiple Laws, or (2) the Law had no title at all. This made it hard to assign a precise label to the Law—however this was still enough for us to classify the Law into one of the four categories that we developed.

Some Concrete Examples of Eigenvector Centrality

For concreteness, in Figure S2 we chose three specific Laws as case studies and tracked their rankings over time. Each Law exemplifies the aforementioned trends in the Law category that it belongs to. The Law "Caught" regulates the most common method of batter dismissal, where the ball is caught after being hit before it touches the ground. While it experiences a period of brief popularity from 1900 to 1923, we see a marked and steady decline in its ranking from 1923 onward. The Law "Fair/Unfair Play" contains rules outlining conventions of the game that protect the "Spirit of Cricket". While this Spirit of Cricket lacks a precise definition, it is explained in the Preamble to the Laws of Cricket Lord's [2025], and contains provisions on issues such as ball tampering and distracting the opposition. The Law was only introduced in the later years of the Laws of Cricket, from 1939 onwards—but ever since its introduction, was consistently one of the highest-ranked Laws. The Law "The Fieldsman" contains rules regarding the fieldsmen, who are the players that try to collect the ball after it is struck by the striking batter, so as to limit the number of runs scored by the striker or to get a batter out. This

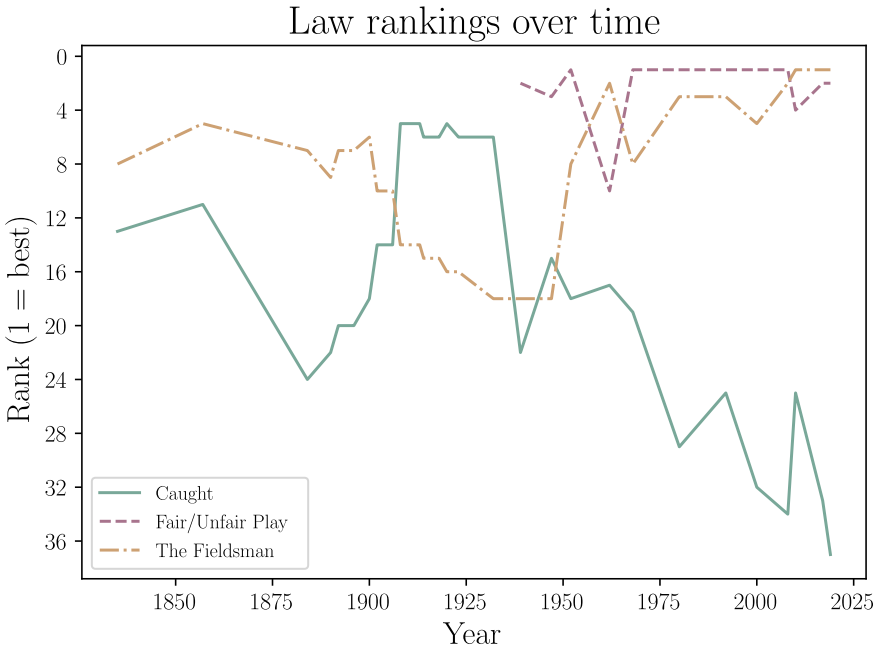

Fig. S2. Ranking of "Caught", "Fair/Unfair Play" and "The Fieldsman" over time. Respectively, these show examples of: Laws that start out prominent but become less relevant in recent times; laws that remain important since their introduction; and laws that fluctuate in their importance over time.

Law ranked fairly highly in the beginning (6th-8th), before dipping in importance from 1902 to 1947. However, it regained ranking thereafter, eventually becoming the most important Law in the game.

Cumulative matches played and time as explanatory variables

In this section, we explain the trade-offs of using cumulative matches played versus time as the explanatory variable, explain how we chose between them for each plot, and point to supplemental figures comparing both sets of plots.

The main drawback of cumulative matches played is that match data is only available from 1877 to 2019: 143 of the 268 years covered by the rules of cricket, and 28 of the 44 rule sets. However, over this period, the two variables can be used effectively interchangeably. Cumulative matches played exhibits an exponential relationship with time, so the trend of the plot for these years would be similar regardless of whether we used time or the logarithm of cumulative matches played as the explanatory variable.

The main advantage of cumulative matches played is that it is more informative as an explanatory variable. It is not the passage of time itself that should drive ruleset change; rather, we hypothesise that events triggering change become more likely as more matches are played. Thus, where possible, we used cumulative matches played as the explanatory variable.

We switched to time in the few cases where the trend before 1877 was informative. In most cases, however, the early period was noisy or monotonous when plotted against time, and obscured the clearer signal post-1877. Since match data is unavailable before 1877, plots against cumulative matches omit this early period entirely. For transparency, we show the full-duration data plotted against time as well: the relevant textual analysis plots Fig. 2(b), (c) and (f)) are in Fig. S12, and the tree structure plots (Fig. 5(b), (d), (e) and (f)) are in Fig. S13.

1 Supporting Figures

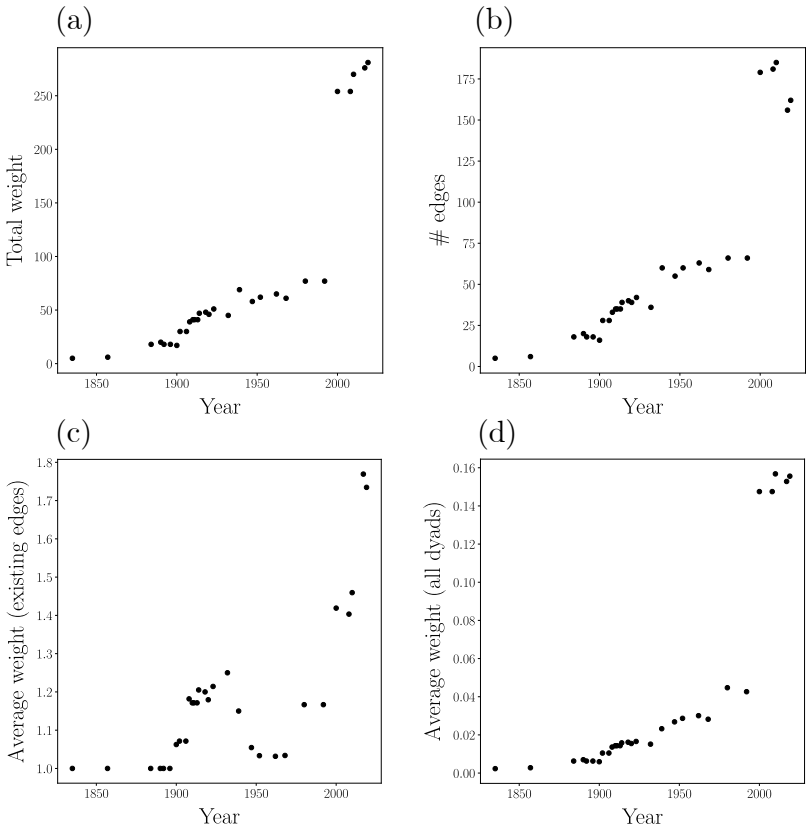

Fig. S3. Descriptive statistics for the interdependency network representations of the cricket rule sets.

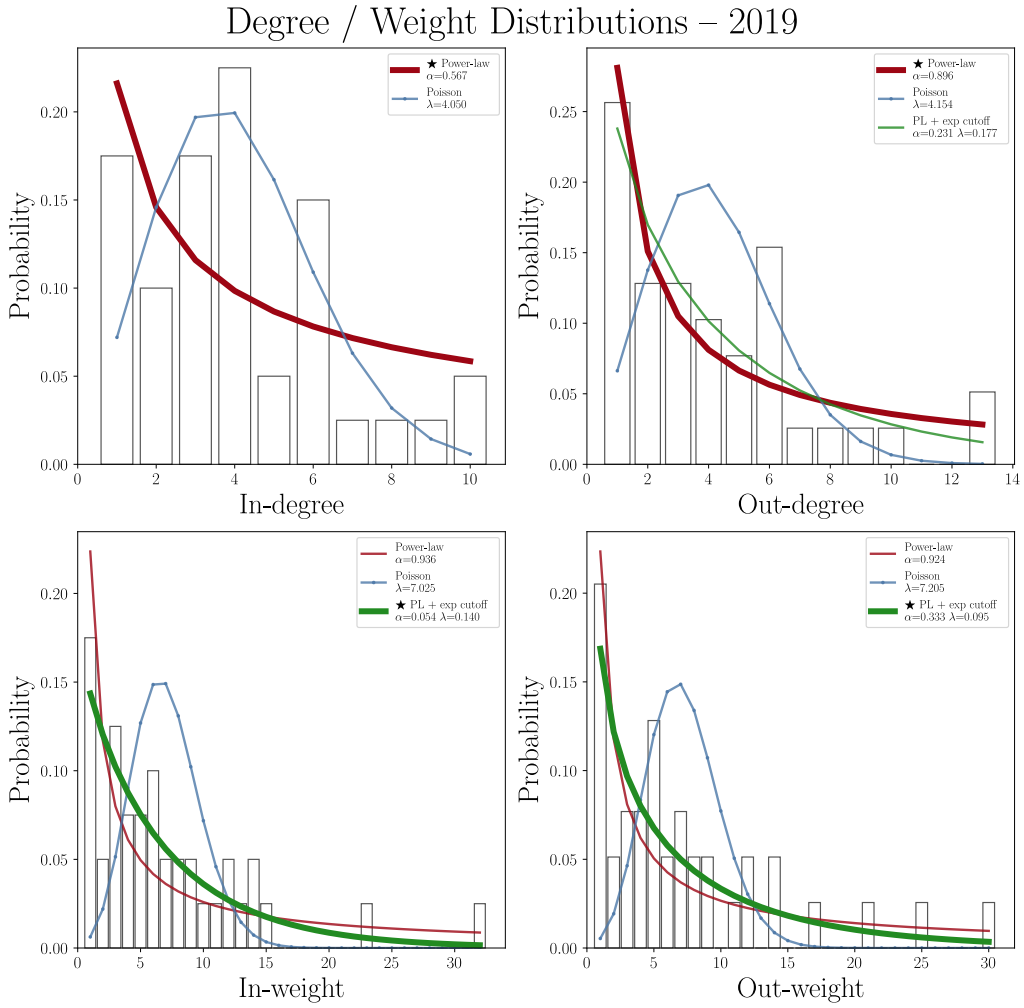

Fig. S4. In-degree, out-degree, in-weight and out-weight distributions for the 2019 cricket rule set's interdependency network.

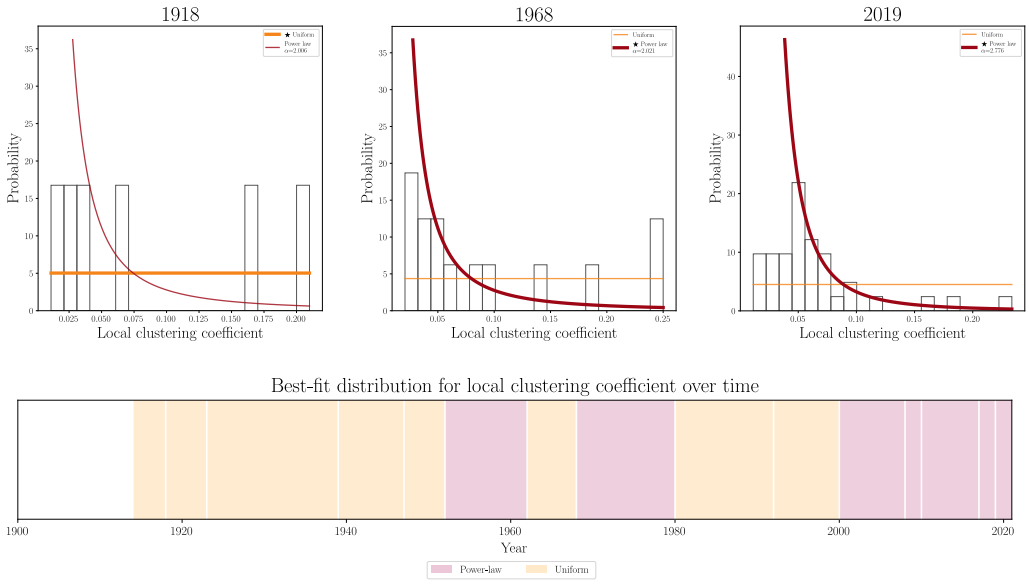

Fig. S5. Clustering coefficient distributions of the interdependency networks [Watts and Strogatz, 1998]. Random and scale-free networks commonly exhibit a uniform clustering coefficient distribution, while hierarchical networks commonly exhibit a power law clustering coefficient distribution. Top row: clustering coefficient distributions for 1918, 1962 and 2019. Bottom: heatmap indicating the best model fit for clustering coefficient distribution over the duration of the rule set. We find consistent evidence that the power-law model is a better fit for the data from 2000 to 2019 [Vuong, 1989]. This transition suggests the emergence of a more hierarchical organization in the interdependency networks over time.

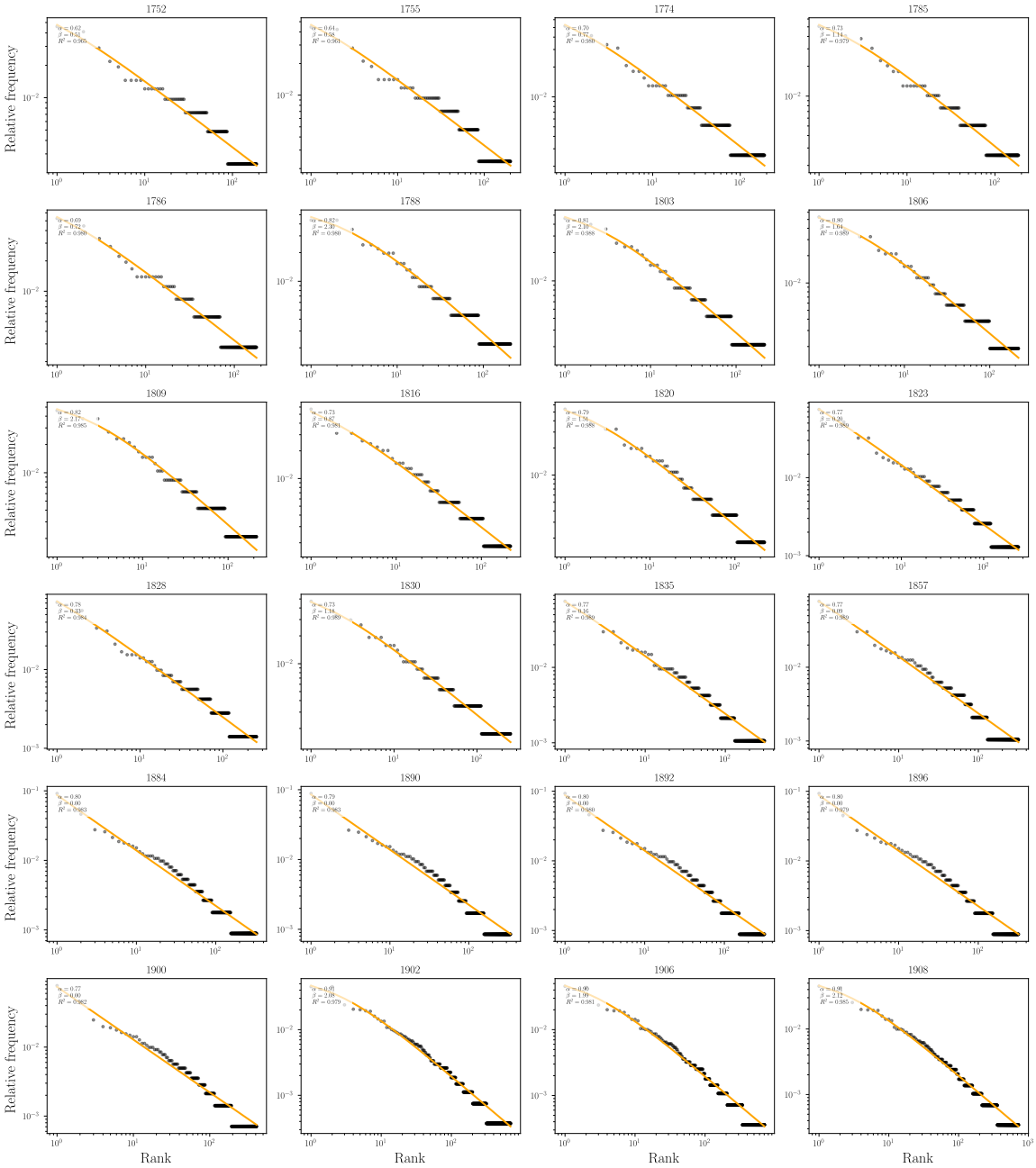

Fig. S6. Zipf-Mandelbrot fits for the rank-frequency distributions of the first 24 rule sets.

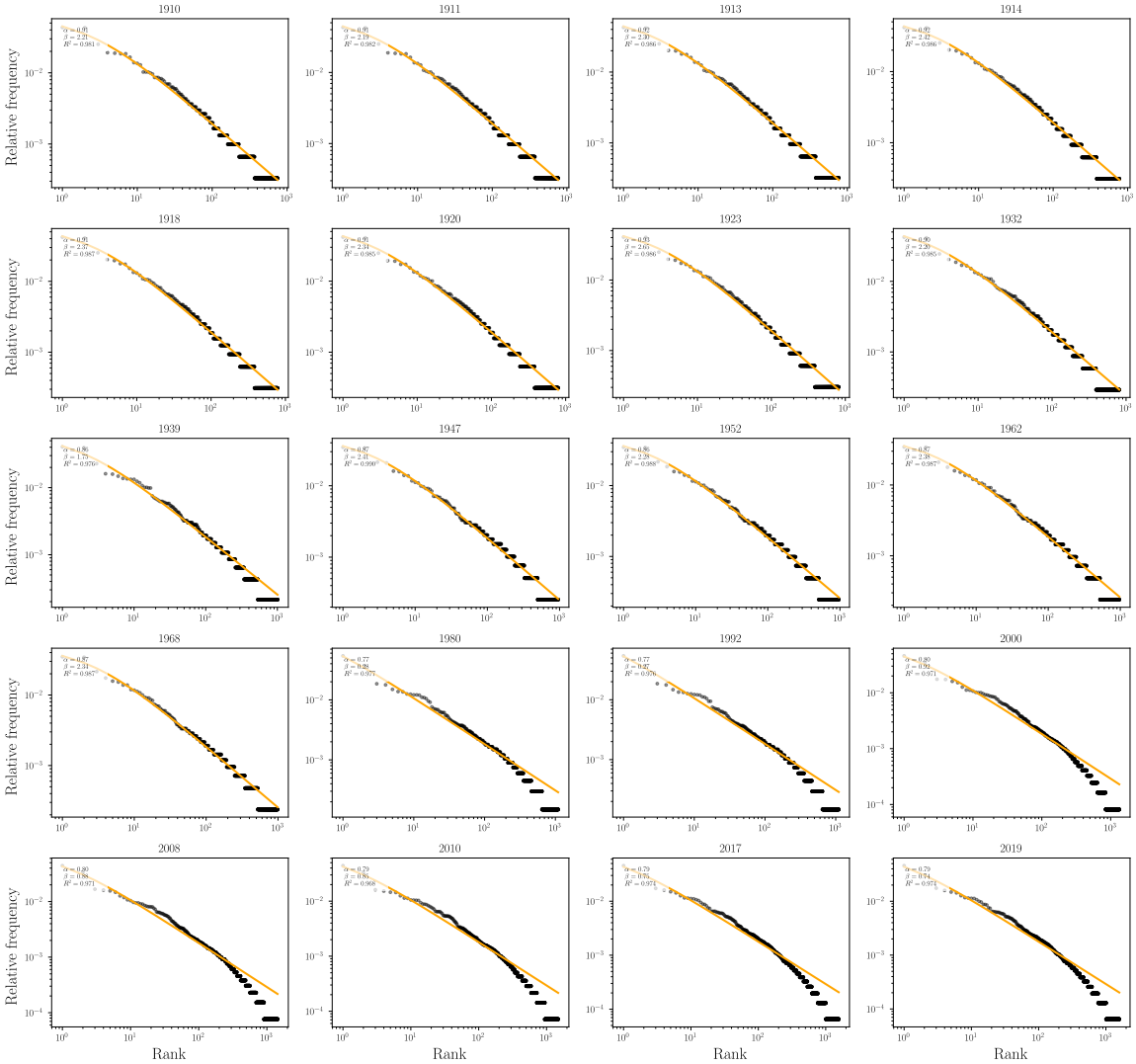

Fig. S7. Zipf-Mandelbrot fits for the rank-frequency distributions of the last 20 rule sets.

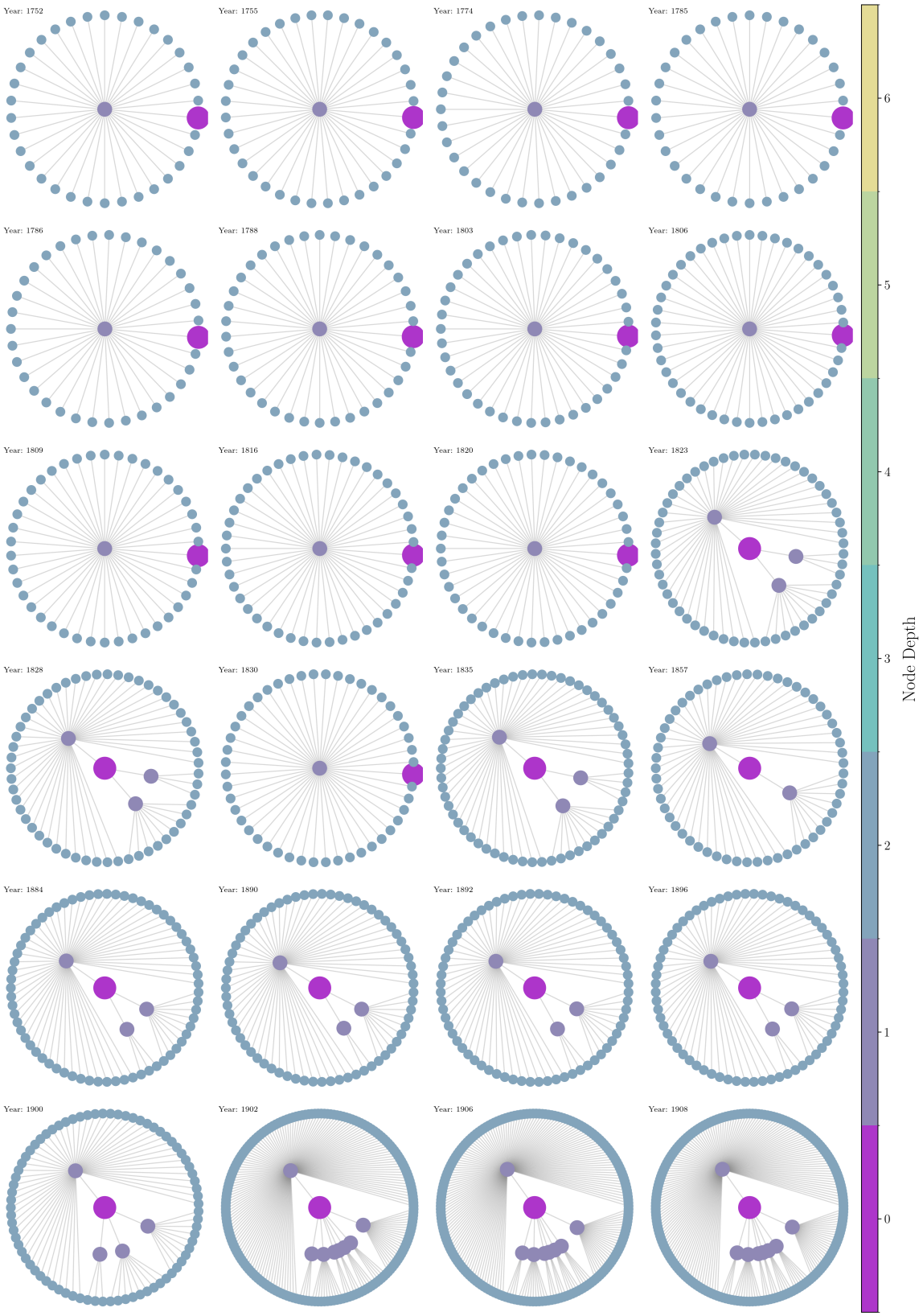

Fig. S8. Rule trees over the first 24 rule sets.

Fig. S9. Rule trees over the last 20 rule sets.

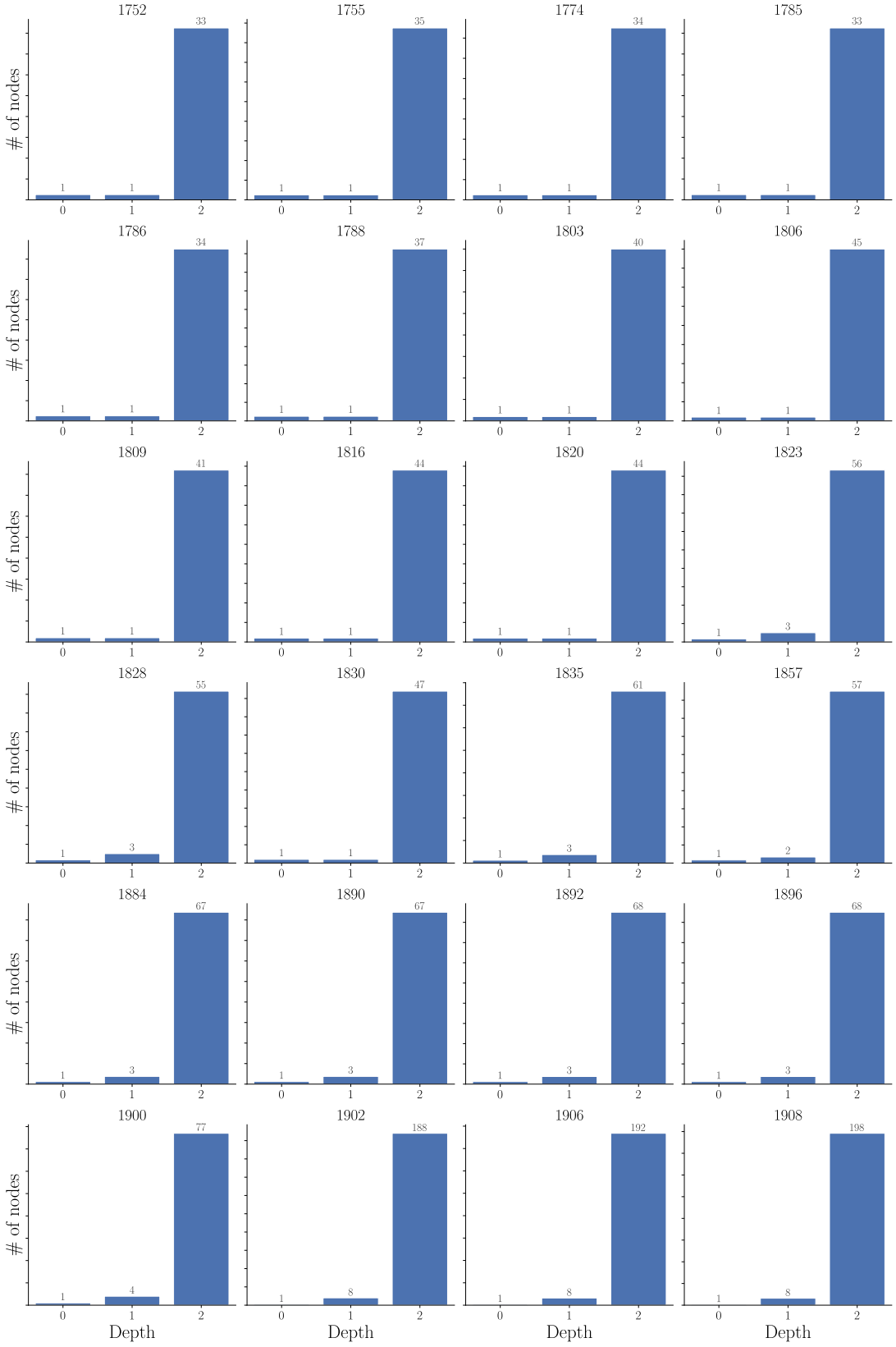

Fig. S10. Depth distributions of the rule trees for the first 24 rule sets.

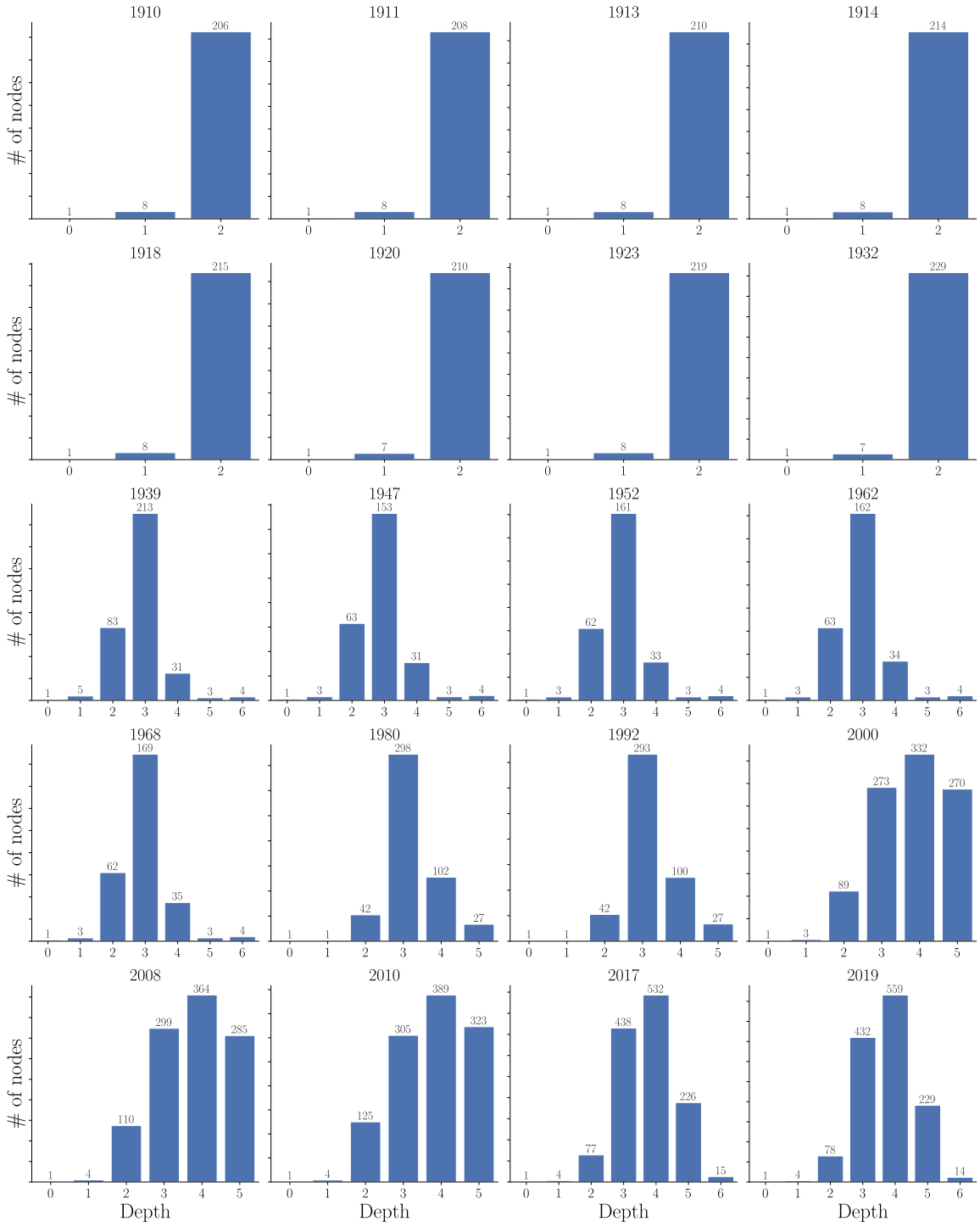

Fig. S11. Depth distributions of the rule trees for the last 20 rule sets.

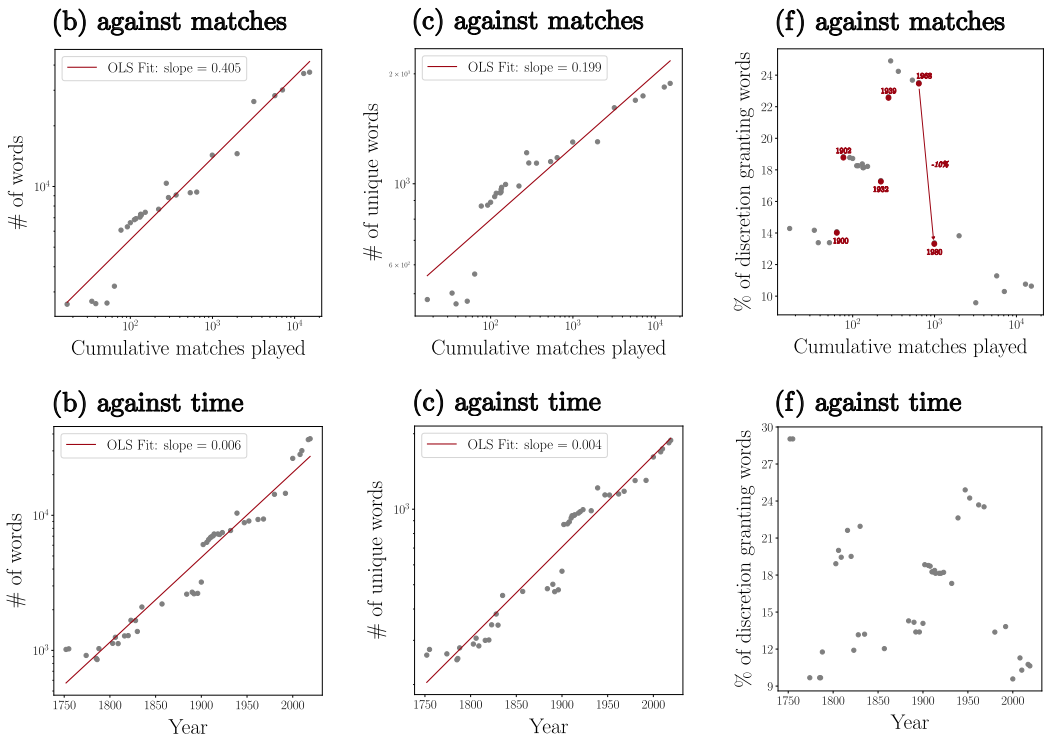

Fig. S12. Comparison of text-based analyses (see Fig. 2(b), (c) and (f)) with cumulative matches played as the explanatory variable (top row) versus with time as the explanatory variable (bottom row). For (b) Number of words in the rule set, regressing against cumulative matches played gives $R^2 = 0.935$ while regressing against time gives $R^2 = 0.948$. For (c) Number of unique words in the rule set, regressing against cumulative matches played gives $R^2 = 0.837$ while regressing against time gives $R^2 = 0.942$.

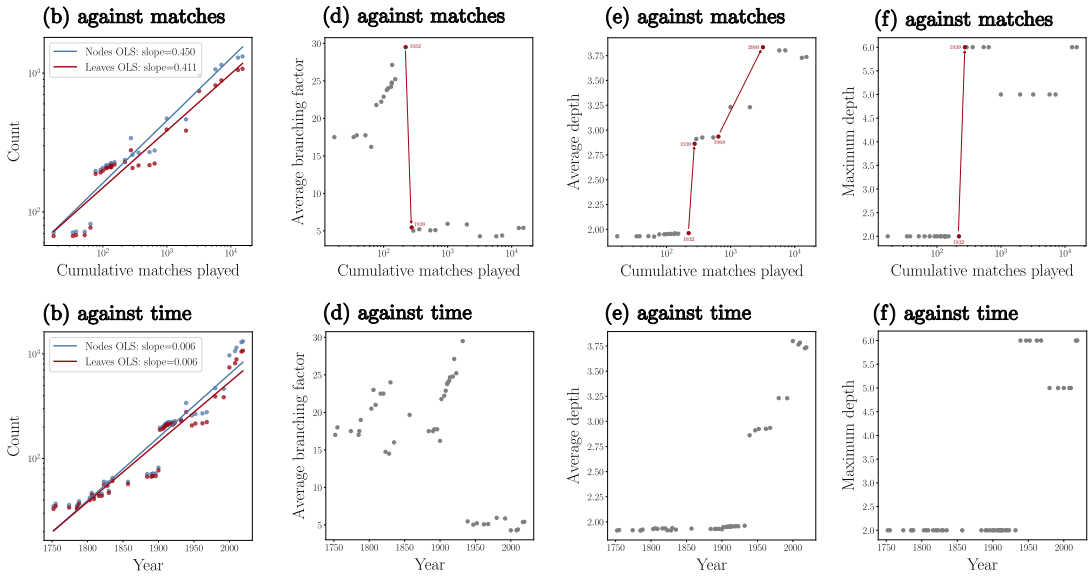

Fig. S13. Comparison of tree-based analyses (see Fig. 5(b), (d), (e) and (f)) with cumulative matches played as the explanatory variable (top row) versus with time as the explanatory variable (bottom row). For (b) Count of nodes and leaves, regressing against cumulative matches played gives $R^2 = 0.918$ for nodes and $R^2 = 0.890$ for leaves, while regressing against time gives $R^2 = 0.911$ for nodes and $R^2 = 0.910$ for leaves.

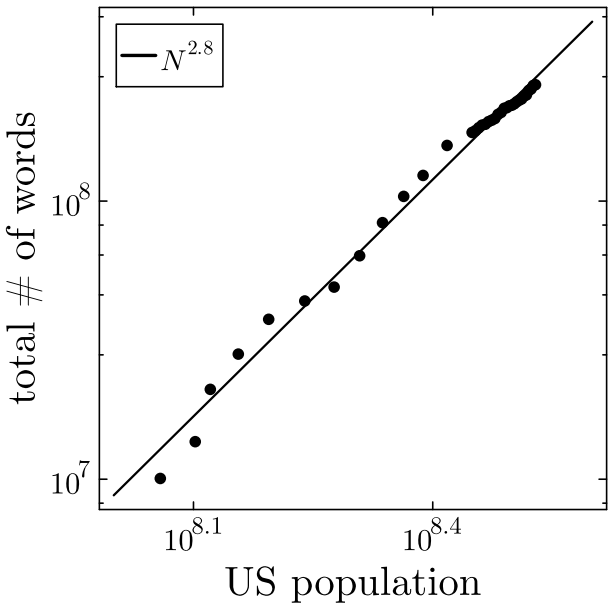

Fig. S14. Plot showing the scaling of the size of the US Code (from data in Jeong et al. [2026]) against population size of the USA.

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